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DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Lab Report No: 01

Course Title: Research Methodology

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Lab Report on: Introduction to Research Methodology

Submitted By	Submitted To
Name: Kuldip Saha Mugdha ID: IT22018 Year: 3 rd Semester: 2 nd Session: 2021-2022 Dept. of ICT, MBSTU	Dr. Md. Badrul Alam Miah Professor Dept. of ICT, MBSTU

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Introduction:

Research methodology is a systematic way to solve research problems by logically adopting various research techniques. It provides scientific principles for planning, designing, and conducting research. A clear understanding of research and methodology is essential for students to conduct academic and practical investigations effectively.

Objective:

The objective of this experiment is to understand the basic concepts of research methodology, including the meaning of research, methodology, and to identify the differences and similarities between a thesis and a project.

1. What is Research?

Research is a systematic, objective, and scientific process of collecting, analyzing, and interpreting data to discover new knowledge or to solve a specific problem. It aims to establish facts, develop theories, and contribute to existing knowledge through careful investigation and experimentation.

2. What is Methodology?

Methodology refers to the structured framework and techniques used to conduct research. It includes the research design, methods of data collection, sampling techniques, and data analysis procedures. Methodology ensures that the research is reliable, valid, and repeatable.

3. Similarities between Thesis and Project:

- Both are based on systematic research methodology.
- Both require defining a research problem and objectives.
- Both involve data collection and analysis.
- Both are supervised and evaluated academically.
- Both require submission of a written report.

4. Distinguish between Thesis and Project:

Criteria	Thesis	Project
Nature	Research-oriented	Application-oriented
Purpose	Creation of new knowledge	Practical implementation
Academic Level	Postgraduate / Doctoral	Undergraduate / Diploma
Duration	Long-term	Short-term
Outcome	Written research document	Report with practical work

Result:

From this experiment, we gained a clear understanding of research methodology and its components. The differences and similarities between thesis and project were successfully identified.

Conclusion:

Research methodology plays a vital role in conducting scientific research. Understanding research, methodology, thesis, and project help students develop analytical thinking and research skills. Both thesis and project work contribute significantly to academic learning and practical experience.